

# Gastrointestinal and endocrine physiology

## Pancreas and Diabetes

Integrated Biomedical and Clinical Sciences OMFS 512

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Mission: To advance the science and art of health through  
education, service, scholarship and social accountability

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# Learning Objectives

## Learning Outcomes

1. Describe the structure of pancreas.
2. Compare and contrast the exocrine and endocrine functions of the pancreas.
3. Describe different types of diabetes
4. Manifestations and complications of diabetes

## Outline

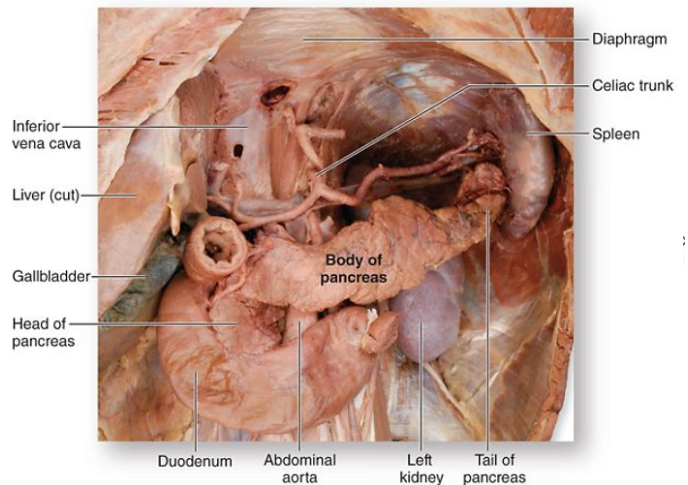
- 1- Exocrine vs Endocrine pancreas.
- 2- Insulin release and function.
- 3- Type 1 diabetes mellitus.
- 4- Type 2 diabetes mellitus.
- 5- Gestational diabetes and diabetes insipidus.

## The pancreas

Pancreas, three anatomical regions:

- a) Head of Pancreas
- b) Body of pancreas
- c) Tail of pancreas.

Begin developing from W4 from a ventral bud and a dorsal bud w. endoderm origin. ancreas begins secreting insulin from W10.

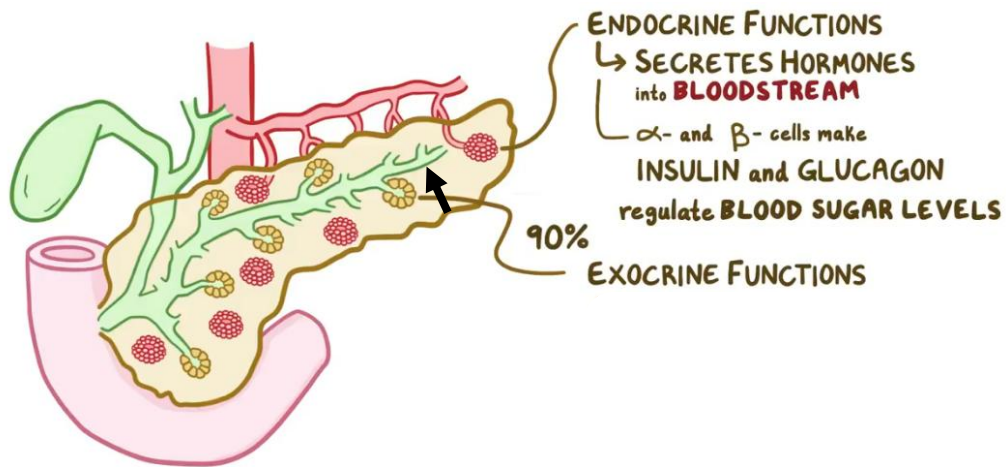


[Pathology of the Pancreas | Pathology: A Modern Case Study, 2e | AccessMedicine | McGraw Hill Medical \(cnsu.edu\)](#)

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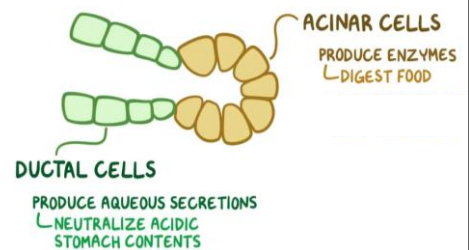
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# Pancreas



## Exocrine Pancreas

- Building block of exocrine pancreas is the acinus. Acini= organized secretory cells around a central lumen.
- Acinar cells are responsible for enzymatic secretions.
  - Synthesize pancreatic enzymes and store the inactive enzymes, known as zymogens, in secretory granules.
- Ductal cells are responsible for the aqueous secretions.



## Acinar Cell Enzymatic Secretions

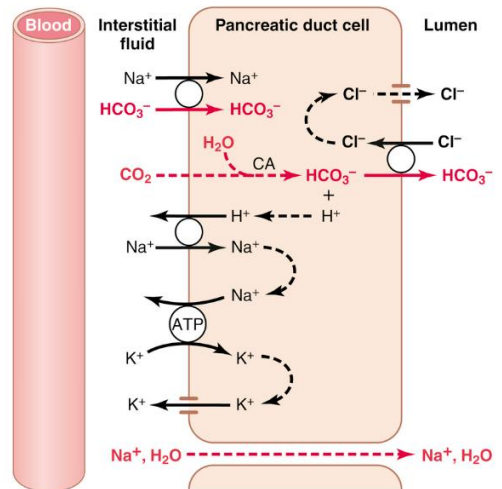
- **For proteins:** Trypsinogen, chymotrypsinogen and procarboxypeptidase (secreted as zymogens).
  - Enterokinase (aka, enteropeptidase): protease found in the duodenum and it activates trypsinogen to yield trypsin.
  - Activated trypsin activates chymotrypsinogen and procarboxypeptidase to yield chymotrypsin and carboxypeptidase.
- **For carbohydrates:** pancreatic amylase, secreted in the active form.
- **For lipids:** pancreatic lipase, secreted in the active form.
- **For nucleic acid:** deoxyribonucleases and ribonucleases are produced to digest the DNA and RNA, respectively.
- Acinar cells also secrete trypsin inhibitor.

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## Ductal Cell Secretions

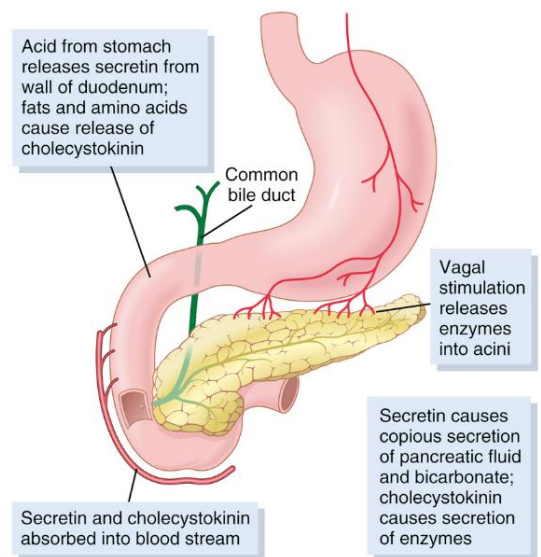
- Aqueous secretions, bicarbonate rich fluid.  
Provide optimal pH for pancreatic enzymes to function & bicarbonate neutralizes gastric acid.
- In ductal secretions, chloride and bicarbonate concentration vary by the pancreatic flow rate.
  - Higher bicarbonate concentrations (and low chloride) are usually observed in high flow rate conditions (stimulated conditions or during active digestion).
- Sodium levels is equivalent to that seen in the plasma.



## Regulation of pancreatic secretions

### Stimuli

- Cephalic phase, vagal stimulation.
- Gastric phase, vagal stimulation and secretin.
- Intestinal phase via CCK and secretin as well as vagal stimulation.
  - Intestinal phase is the strongest stimulator for pancreatic secretions.
  - CCK is the major stimulator for enzyme-rich pancreatic secretions. Secretin stimulates water and bicarbonate.



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### Multiplicative Effects

Different stimuli = total secretion > sum of the secretions caused by each one stimulus separately. Pancreatic secretion = combined effects of the multiple basic stimuli, not from one alone.

### Inhibitor:

When food reaches the ileum, the neuroendocrine cells will secrete the hormone **peptide YY** which will decrease pancreatic secretions.

## Endocrine Pancreas

### 5 DIFFERENT CELL TYPES

\* SECRETE SPECIFIC HORMONES

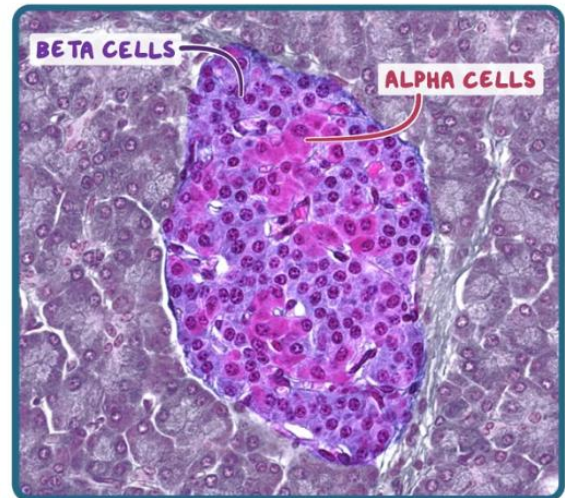
**BETA (ABUNDANT) → INSULIN**

**ALPHA → GLUCAGON**

**DELTA → SOMATOSTATIN**

**GAMMA → PANCREATIC  
POLYPEPTIDE**

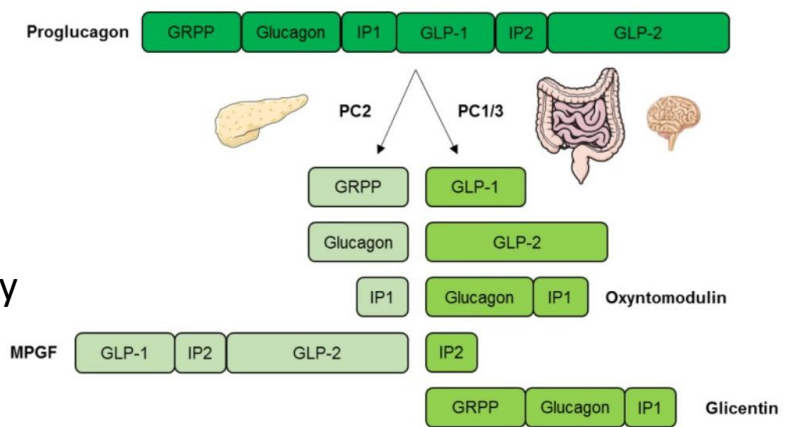
**EPSILON → GHRELIN**



GOMORI'S STAIN

# Glucagon

Peptide hormone synthesized by  $\alpha$  cells of pancreas as proglucagon. Then, proglucagon is processed into glucagon by prohormone convertase 2 (PC2).

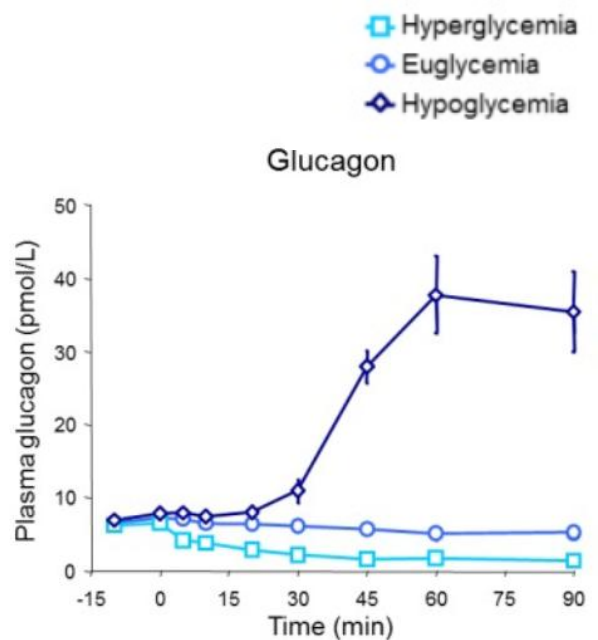


### Physiological role:

Catabolic hormone “elevates blood sugar levels to glucose homeostasis”.

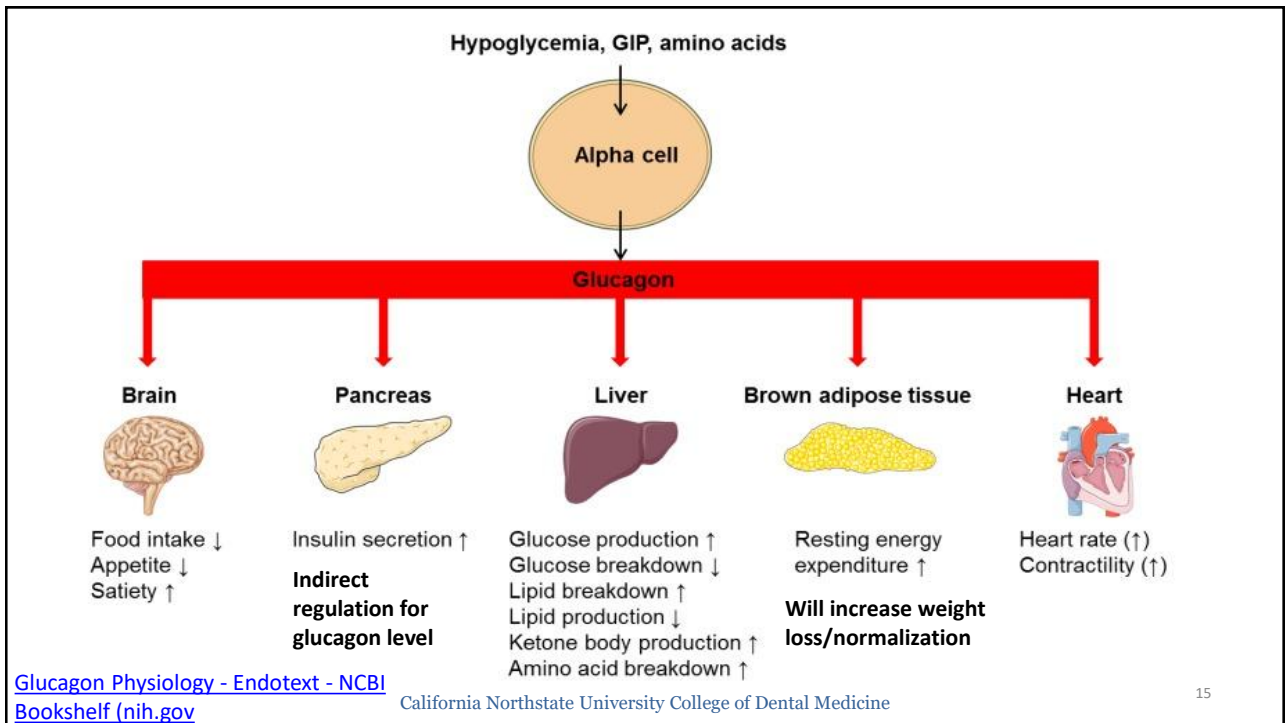
### Regulation:

- Stimuli: **hypoglycemia**, exercise, norepinephrine and acetylcholine.
- Inhibitors: Inhibited by insulin, somatostatin, amylin.



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**Somatostatin:** secreted by the delta cells of the pancreas. Inhibits the secretion of insulin, glucagon, and gastrin.

**Pancreatic polypeptide:** Secreted by gamma (PP) cells. Regulates the endocrine and exocrine function of the pancreas.

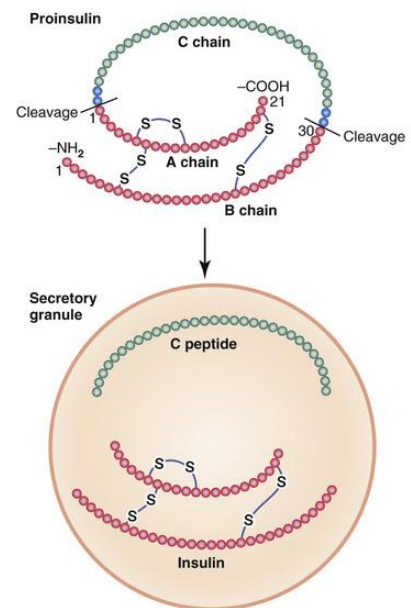
**Ghrelin:** Secreted by epsilon cells. It stimulates appetite “hunger hormone” and stimulate the release of growth hormone.

**Amylin** stored in  $\beta$ -cell granules and released along with insulin. Plasma amylin concentrations increase after a meal or glucose infusion. It helps in suppressing the postprandial secretion of glucagon and slow gastric emptying.

## Insulin: Biosynthesis

Peptide hormone synthesized by  $\beta$  cells (chromosome 11).

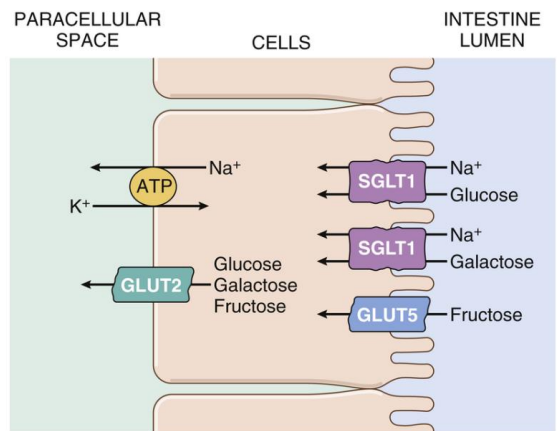
- Synthesized as inactive proinsulin in  $\beta$  cells.
- Proinsulin consists of three peptide chains A and B & connecting peptide (C peptide).  
Detecting C-peptide level = endogenous insulin level.
- Insulin is formed of two amino acid chains (A and B) connected to each other by disulfide linkage and stored in secretory granules.



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Following a meal, Sodium dependent transporter “SGLT-1” transfers glucose from the intestinal lumen into the intestinal mucosa.

Then, GLUcose Transporter 2 (sodium independent, GLUT-2) transport glucose through the cell’s membrane into the paracellular space and then, blood



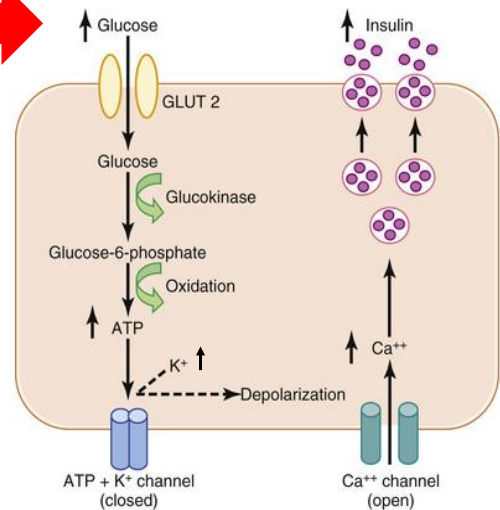
## Mechanism of Insulin Release

Following a Meal

Pancreas, following a meal:

- GLUT2 transport glucose into beta cells= increase of ATP/ADP ratio.

This increases the intercellular calcium concentration, fusion of insulin secretory granules with the plasma membrane and insulin release.



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## Regulation of Insulin Release

### Factors that Increase Insulin Secretion

↑ blood glucose

↑ amino acids (arginine, lysine, leucine)

↑ fatty acids

Glucagon

GIP

ACh

### Factors that Decrease Insulin Secretion

↓ blood glucose

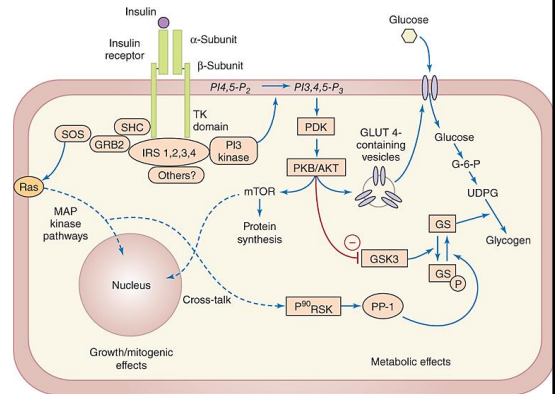
Somatostatin

Norepinephrine, epinephrine

## Insulin action

**Anabolic hormone with important role in carbohydrate, lipid and protein metabolism.**

Insulin activate insulin receptors (IR) leading to the phosphorylation (activation) of signaling proteins (like, insulin receptor substrates IRS), translocation of glucose transporter (GLUT-4) to plasma membrane surface, enhanced glucose entry, glycolysis, glycogen synthesis, and regulation of protein synthesis.



Source: David G. Gardner, Dolores Shoback: Greenspan's Basic & Clinical Endocrinology, Tenth Edition Copyright © McGraw-Hill Education. All rights reserved.

# Glucose transporters

| Transporter                                                                                 | Expression                                                                                                                                                                                                                                                                                 | Function                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GLUT-1     | Ubiquitous, with particularly high levels in human erythrocytes and in the endothelial cells lining the blood vessels of the brain. Expressed in skeletal muscle and fat                                                                                                                   | Glucose uptake by skeletal muscle and fat under basal conditions                                                                                                         |
| GLUT-2    | Low-affinity glucose transporter present in pancreatic $\beta$ -cells, liver, intestine, and kidney                                                                                                                                                                                        | Functions in the glucose sensor system and ensures that glucose uptake by pancreatic $\beta$ -cells and hepatocytes occurs only when circulating glucose levels are high |
| GLUT-3   | Primarily in neurons                                                                                                                                                                                                                                                                       | Together, GLUT-1 and GLUT-3 are crucial in allowing glucose to cross the blood–brain barrier and enter neurons                                                           |
| GLUT-4  | Predominantly in striated muscle and adipose tissue. In contrast to the other GLUT isoforms, which are primarily localized on the cell membrane, GLUT-4 transporter proteins are sequestered in specialized storage vesicles that remain within the cell's interior under basal conditions | The major insulin-responsive transporter                                                                                                                                 |

## Insulin action and fate

- Insulin-stimulated glucose uptake is mainly mediated by GLUT-4.
  - Number insulin receptors is modulated by exercise, diet, and several hormones, e.g, exercise upregulates the number of receptors.
  - Chronic exposure to high insulin levels, obesity, and excess growth hormone can lead to a downregulation of insulin receptors (insulin resistance “like type 2 diabetes”).
- Following secretion, its plasma half-life around 5 minutes.
- Secreted and non-receptor bound insulin is cleared from circulation within 10 minutes via insulinase enzyme (mostly in liver; kidneys and muscles “low”).

## Physiologic Functions of Insulin

| Metabolic Effects       | Insulin Stimulates                                                                                                                                                         | Insulin Inhibits                                                                                                             |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Carbohydrate metabolism | Glucose transport in adipose tissue and muscle<br>Rate of glycolysis in muscle and adipose tissue<br>Glycogen synthesis in adipose tissue, muscle, and liver               | Glycogen breakdown in muscle and liver<br>Gluconeogenesis in the liver                                                       |
| Lipid metabolism        | Fatty acid and triacylglycerol synthesis in tissues<br>Uptake of triglycerides from the blood into adipose tissue and muscle<br>Rate of cholesterol synthesis in the liver | Lipolysis in adipose tissue, lowering the plasma fatty acid level<br>Fatty acid oxidation in muscle and liver<br>Ketogenesis |
| Protein metabolism      | Amino acid transport into tissues<br>Protein synthesis in muscle, adipose tissue, liver, and other tissues                                                                 | Protein degradation in muscle<br>Urea formation                                                                              |

# Disorders of the Pancreas

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## Pancreatitis

**Causes:** alcoholism (common cause in adults). Cystic fibrosis (common cause in children).

- **Acute Pancreatitis:** Caused by activation of pancreatic enzymes leading to pancreatic autodigestion. Patient present with sudden-onset epigastric abdominal pain that radiates to the back accompanied with anorexia and nausea.
- **Chronic Pancreatitis:** repeated episodes of acute inflammation or pancreatitis lead to fibrosis and damage. In chronic pancreatitis, the sudden epigastric pain is often absent. Symptoms include persistent pain, worsened by fatty meals or alcohol.

**Complications:**

- Pancreatic insufficiency leading to steatorrhea
- weight loss, and deficiencies of fat-soluble vitamins (A, D, E, and K).
- Long term complication: diabetes and may cause pancreatic cancer.

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# Diabetes

HOT & DRY = SUGAR HIGH



COLD & GLAMMY =  
NEED SOME CANDY

 AMERICAN DIABETES ASSOCIATION

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## Fast Facts on Diabetes

### Diabetes

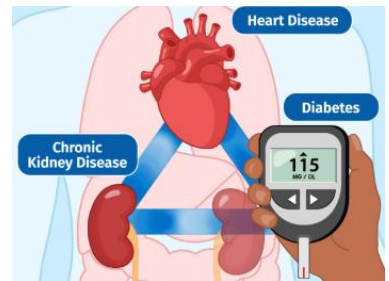
- **Total:** 38.4 million people have diabetes (11.6% of the US population)
- **Diagnosed:** 29.7 million people, including 29.4 million adults
- **Undiagnosed:** 8.7 million people (22.8% of adults are undiagnosed)

### Prediabetes

- **Total:** 97.6 million people aged 18 years or older have prediabetes (38.0% of the adult US population)
- **65 years or older:** 27.2 million people aged 65 years or older (48.8%) have prediabetes

## Screening

Diabetes is associated with increased risks of cardiovascular disease (CVD), kidney failure and nonalcoholic fatty liver disease.



Therefore,

*“Screening asymptomatic adults for prediabetes and type 2 diabetes may allow earlier detection, diagnosis, and treatment, with the ultimate goal of improving health outcomes.”*

[Recommendation: Prediabetes and Type 2 Diabetes: Screening](#)  
[| United States Preventive Services Taskforce](#)  
[\(uspreventiveservicestaskforce.org\)](#)

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## Diagnostic Criteria for Diabetes Mellitus

1.  $HbA_{1c}$  (as measured in a DCCT-referenced assay)  $\geq 6.5\%$   
OR
2. FPG  $\geq 126$  mg/dL (7.0 mmol/L); fasting is defined as no caloric intake for at least 8 hr  
OR
3. 2-hr plasma glucose  $\geq 200$  mg/dL (11.1 mmol/L) during an OGTT  
OR
4. In a patient with classic symptoms of hyperglycemia or hyperglycemic crisis, a random plasma glucose  $\geq 200$  mg/dL (11.1 mmol/L)

From American Diabetes Association (ADA): 2016 guidelines. Available at <http://www.ndei.org/ADA-diabetes-management-guidelines-diagnosis-A1C-testing.aspx>.

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**Glycated hemoglobin** (A1C, also termed hemoglobin A1C or HbA1c). Physiologically, around 3 % of hemoglobin in circulating RBCs is glycated during their lifespan.

HbA1 C levels are used to diagnose diabetes and to monitor the efficacy of treatment.

- Normal A1C  $\leq$  5.6 %.
- Diabetic A1C  $\geq$  6.5 % & Prediabetic A1C: between 5.7 to 6.4 %.
- A1C can be falsely low, caused by: alcohol use, pregnancy, bleeding, anemia, kidney disorder.
- A1C can be falsely high, caused by: vit. B12 deficiency, organ transplant, immunosuppressants



### **Fasting plasma glucose (FPG)**

calculated by measure the plasma concentration of glucose following an eight-hour fasting.

- Normal FPG: 70 - 100 mg/dL.
- Diabetic FBG  $\geq$  126 mg/dL

### **A non-fasting or Random glucose test:**

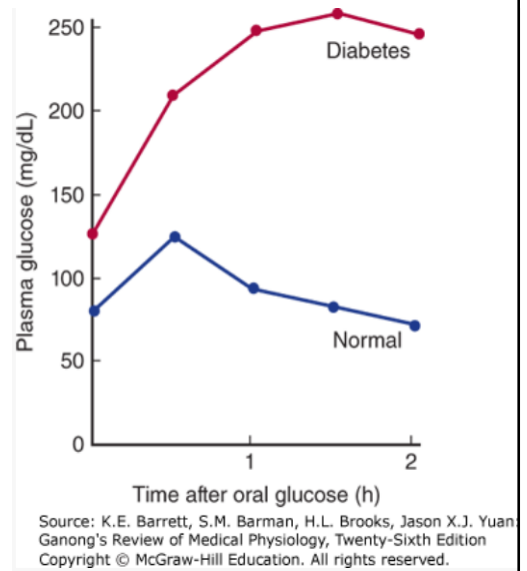
Performed at any time, 200 mg/dL or higher is diagnostic for diabetes as long as the patient is displaying symptoms.

- Normal levels <140 mg/dL.

## Two-hour oral glucose tolerance test (OGTT).

This test measures the plasma glucose level before and at different time points following the ingestion of oral glucose solution containing the equivalent of 75 g anhydrous glucose dissolved in 300 ml water.

- Normal OGTT < 140 mg/dL
- Diabetic OGTT ≥ 200 mg/dL



### Oral glucose tolerance test



## Type 1 Diabetes Mellitus

**Etiology:** Type 1 diabetes mellitus (T1DM) is caused autoimmune destruction of  $\beta$ -cells. **Insulin deficiency.**

**Prevalence:** Less common, 10 % of diabetic patients.

**Risk factors:** A combination of genetic and environmental predisposing factors can increase the risk for the development of type 1 diabetes.

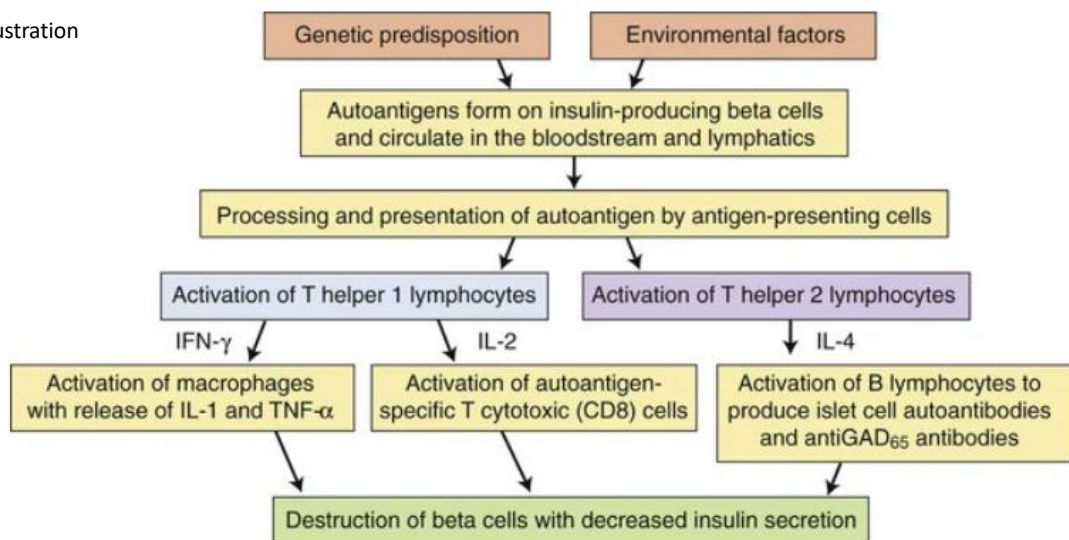
**A) Genetic factors:**

Mutations in specific region of *HLA* gene (*HLA-DR3* and *DR4*) contribute to 40–50% of the heritable risk for type 1 diabetes.

**B) Environmental risk factors** can act as “accelerators” of beta cell autoimmunity. These include viral infections, infant nutrition and hygiene.

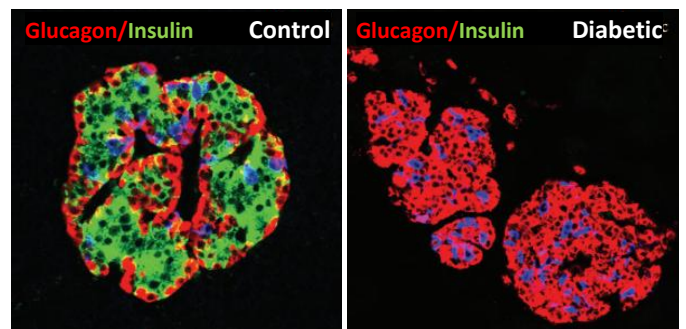
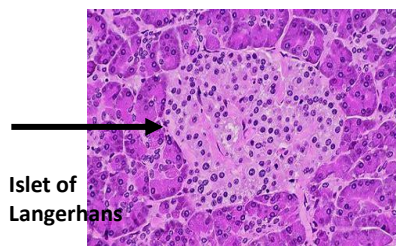
For example, viral infection (enteroviruses, especially Coxsackie virus B [CVB]) is associated w. increased risk of type 1 diabetes.

For illustration



**FIGURE 22.12** Pathophysiology of Type 1 Diabetes Mellitus. *GAD<sub>65</sub>*, Glutamic acid decarboxylase; *IFN-γ*, interferon-gamma; *IL*, interleukin; *TNF-α*, tumor necrosis factor-alpha.

**Histology:** Absence of insulin immunostaining due to  $\beta$ -cells destruction and leukocyte infiltration of islets.



Control: Insulin-secreting beta cells are the most abundant in the middle of islet (green staining). Glucagon-secreting alpha cells are found on the periphery (red staining).

**Onset:** Younger age (< 30 Years old). Peak diagnosis age is around 13-14 years old with possible earlier and later diagnosis ages.

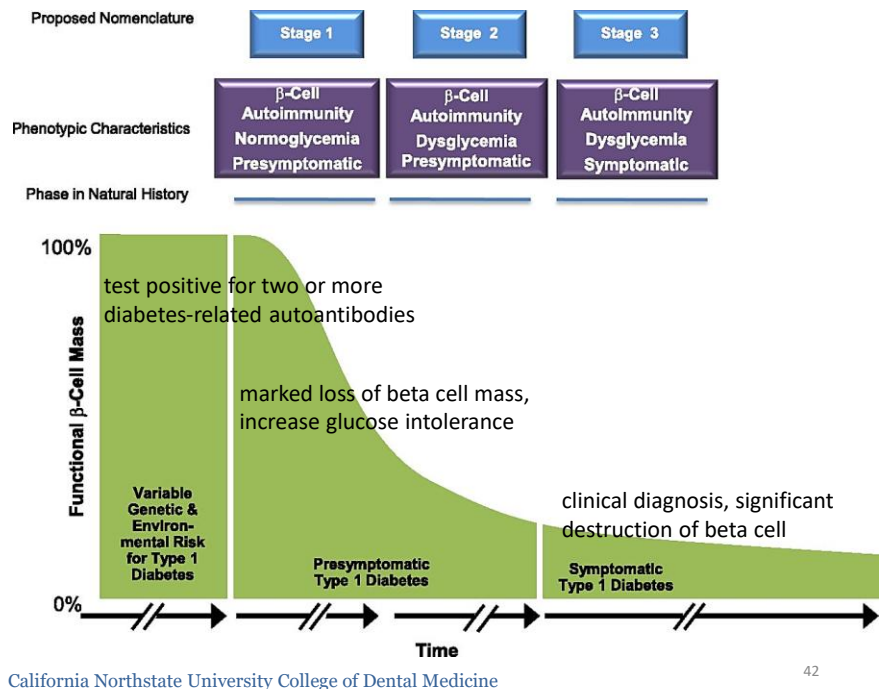
### Pathophysiology

- Diabetes symptoms becomes obvious when 80% to 90% of the insulin-secreting beta cells are destroyed.
- Hypoinsulinemia leads to a marked increase in glucagon secretion.
- Decreased amylin, another  $\beta$ -cell hormone, leads to the increase of glucagon levels.
- Glucagon increases blood glucose level by stimulating glycogenolysis and gluconeogenesis.



## Staging system for type 1 diabetes

[Staging Presymptomatic Type 1 Diabetes: A Scientific Statement of JDRF, the Endocrine Society, and the American Diabetes Association | Diabetes Care | American Diabetes Association \(diabetesjournals.org\)](#)



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## Clinical Symptoms of T1DM

| MANIFESTATIONS | RATIONALE                                                                                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Polydipsia     | Because of elevated blood glucose levels, water is osmotically attracted from body cells, resulting in intracellular dehydration and hypothalamic stimulation of thirst                                                                               |
| Polyuria       | Hyperglycemia acts as an osmotic diuretic; the amount of glucose filtered by the glomeruli of the kidneys exceeds the amount that can be reabsorbed by the renal tubules; glycosuria results, accompanied by large amounts of water lost in the urine |
| Polyphagia     | Depletion of cellular stores of carbohydrates, fats, and protein results in cellular starvation and a corresponding increase in hunger                                                                                                                |
| Weight loss    | Weight loss occurs because of fluid loss in osmotic diuresis and the loss of body tissue as fat and proteins are used for energy as a result of the effects of insulin deficiency                                                                     |
| Fatigue        | Metabolic changes result in poor use of food products, contributing to lethargy and fatigue; sleep loss from severe nocturia also contributes to fatigue                                                                                              |



- \* **BODY MAKES INSULIN**  
but **TISSUES DON'T RESPOND**
- \* **REASON ISN'T FULLY UNDERSTOOD**

## Type 2 Diabetes Mellitus

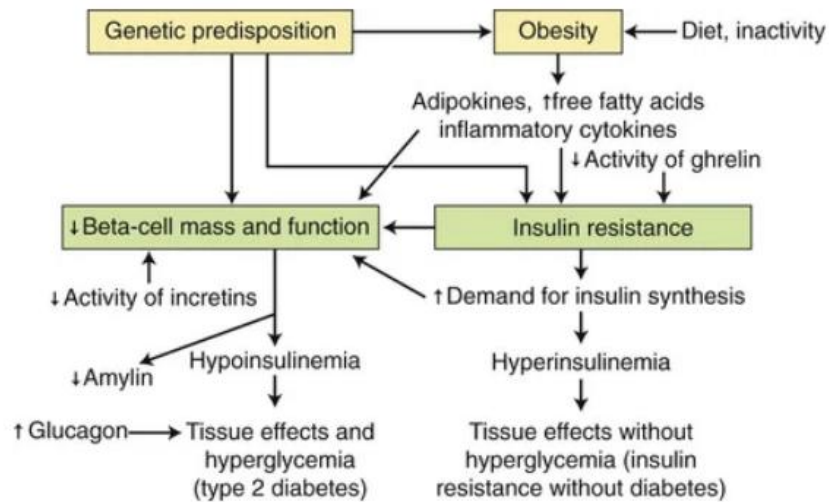
**Etiology:** Multifactorial: a combination of genetic susceptibility, insulin resistance, obesity (metabolic syndrome) and inadequate insulin secretion by the pancreas.

**Prevalence:** Affect around 90 % of the diabetic population

**Onset:** Older age > 45 years old

**Risk Factors:** Prediabetic, overweight, age 45 years or older, positive family history, sedentary lifestyle, ethnicity and history of gestational diabetes (for females).

# Pathophysiology



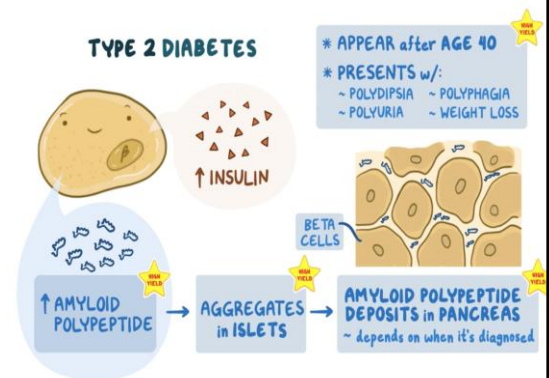
**FIGURE 22.14** Pathophysiology of Type 2 Diabetes Mellitus.

## Two-stage process

### 1- Insulin resistance:

Subnormal cellular response to the normal insulin concentration (either endogenous or exogenous). Abnormal glucose metabolism, fatty liver, skin pigmentation, inflammation, increase risk for CKD, CVS and PCOS.

Pancreas upregulates insulin production to match the hyperglycemic load. At the same time, amyloid polypeptide builds up and aggregates in the islets.



Compensatory hyperinsulinemia prevents the clinical appearance of diabetes for many years.

## **2- Decreased $\beta$ cell mass:**

Following overburdening of the beta cell mass, diabetic patients are presented with a decreased beta cell function over time leading to a relative deficiency of insulin.

## Clinical manifestations

Clinical manifestations of type 2 diabetes are often nonspecific, especially in obese patients. Affected individual is often overweight, hyperinsulinemic, and may be hypertensive.

- fatigue,
- pruritus (itching),
- recurrent infections,
- visual changes, or neuropathy-related symptoms (paresthesia, tingling or prickling).
- Acanthosis nigricans: brownish pigmentation in the skin (back of the neck, armpits) due to insulin resistance and insulin-mediated IGF-1 activation in keratinocytes
- Polyuria, polydipsia, weight loss



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e Copyright © McGraw Hill. All rights reserved.

## Management strategies

- Type I diabetes
  - Exogenous insulin to compensate for endogenous insulin deficiency.
- Type 2 diabetes:
  - First line: diet and exercise to attain a healthy body weight.
  - Combination of anti-hyperglycemic medications

# Acute and Chronic Complications

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# Hypoglycemia

Blood glucose < 70 mg/dL

## HYPOGLYCEMIA — IRREVERSIBLE DAMAGE

### \* ASSOCIATED w/ TREATMENT of DIABETES

- ~ INSULIN
- ~ SULFONYLUREAS



\* Alcohol consumption in the evening greatly increase the likelihood of hypoglycemia the following morning

## SYMPTOMS

### \* SYMPTOMS of AUTONOMIC SYSTEM ACTIVATION

- ~ PALPITATIONS
- ~ SWEATING
- ~ SHAKING
- ~ NAUSEA
- ~ HUNGER



### \* NEUROGLYCOPENIC SYMPTOMS

- ~ FOCAL NEUROLOGICAL DEFICIT
- ~ MENTAL CONFUSION
- ~ COMA



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## Diabetic Ketoacidosis (DKA)

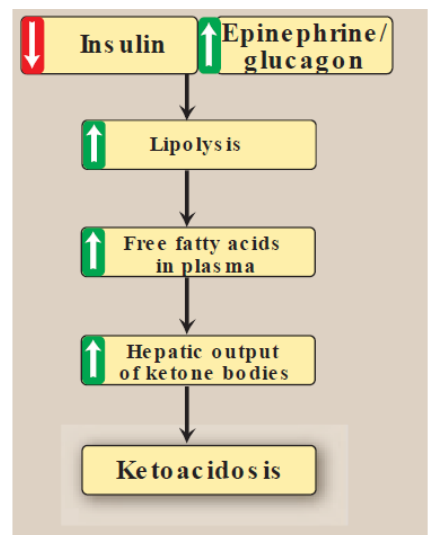
**Acute complication associated with type 1 diabetes.**

Medical emergency that can develop in less than 24 hours.

**Pathophysiology:** From insulin deficiency combined with increased glucagon levels.

Decreased insulin to glucagon ratio promotes lipolysis and excess fatty acid production. Fatty acids are converted to ketone bodies in the liver (like acetoacetic acid and acetone). Accumulation of acidic ketone bodies increase the blood acidity = development of metabolic ketoacidosis.

*Acetone is responsible for the fruity smell from the diabetic person's breath.*



[Diabetes Mellitus: Management and Therapies | Harrison's Principles of Internal Medicine 21e | AccessMedicine | McGraw Hill Medical \(cnsu.edu\)](#) [California Northstate University College of Dental Medicine](#)

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### Symptoms

- Nausea/vomiting
- Thirst/polyuria
- Abdominal pain
- Shortness of breath (aka, kussmaul breathing)

### Precipitating events

- Inadequate insulin administration
- Infection (pneumonia/UTI/gastroenteritis/sepsis)
- Infarction (cerebral, coronary, mesenteric, peripheral)
- Pancreatitis
- Drugs (cocaine)
- Pregnancy

### Management:

IV fluids to replenish the lost fluids, administration of insulin to promote normal metabolic pathways & electrolyte monitoring.

DKA severity depend on several factors including the rate of ketoacid production, metabolism rate of ketoacids, and the rate of acid excretion in the urine.

## Ketosis vs Diabetic Ketoacidosis

**Diabetic ketoacidosis** life-threatening medical condition that arises because of severe hyperglycemia in the absence or deficiency of insulin and requires hospitalization (ketonemia  $>1.5$  mmol/L, ketonuria and hyperglycemia ).

**Ketosis (aka, nutritional ketosis)** refers to an increase in ketone bodies concentration. Occurs in persons consuming a keto or low carbohydrate diet, or a fasting person. In this case, body depends on fat metabolism as the primary source of energy. Further, metabolic needs and demands of the body are balanced (ketone body 0.5-2 mmol/L).

## Hyperosmolar hyperglycemic nonketotic syndrome

HHNKS is also termed hyperglycemic hyperosmolar state (HHS).

**Pathophysiology:** enough insulin level to prevent excessive lipolysis but not enough to promote proper glucose utilization. Patients do not progress to diabetic ketoacidosis.

Hyperglycemia increases the plasma osmolarity leading to glycosuria (excretion of glucose in urine) and polyuria. Causing severe volume depletion, dehydration, and loss of electrolytes.

The hyperglycemic hyperosmolar state is accompanied by neurologic changes that can proceed to coma and even death.

**Prevalence:** Associated with type 2 diabetes. More often in elderly individuals with existing comorbidities.

**Precipitating events:** Incompliance with treatment, infections

**Onset:** few days-few weeks

**Symptoms:** Mental status changes like partial and complete lack of consciousness that can progress to seizures, coma (named hyperosmolar hyperglycemic coma) and even death.

# Long-term Complications

## Microvascular

### Eye

High blood glucose and high blood pressure can damage eye blood vessels, causing retinopathy, cataracts and glaucoma

### Kidney

High blood pressure damages small blood vessels and excess blood glucose overworks the kidneys, resulting in nephropathy.

### Neuropathy

Hyperglycemia damages nerves in the peripheral nervous system. This may result in pain and/or numbness. Feet wounds may go undetected, get infected and lead to gangrene.

## Macrovascular

### Brain

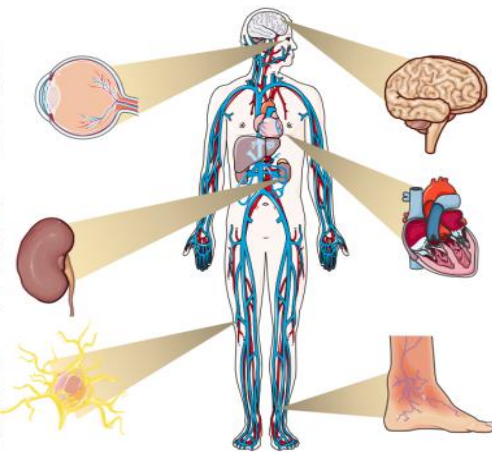
Increased risk of stroke and cerebrovascular disease, including transient ischemic attack, cognitive impairment, etc.

### Heart

High blood pressure and insulin resistance increase risk of coronary heart disease

### Extremities

Peripheral vascular disease results from narrowing of blood vessels increasing the risk for reduced or lack of blood flow in legs. Feet wounds are likely to heal slowly contributing to gangrene and other complications.



Hyperglycemia increases the risk for infections as well as the morbidity and mortality from infectious agents. This could be attributed to

- a) Hypoxia: glycosylated hemoglobin hinders oxygen release to the tissues.
- b) Decreased blood supply to peripheral tissues secondary to vascular changes.
- c) Recurrent infections. Pathogens can proliferate better in hyperglycemic conditions because glucose is an excellent source of energy.
- d) Chronic hyperglycemia can impair both innate and adaptive immune responses.
- e) Delayed wound healing due to slower collagen synthesis and decreased angiogenesis.

## Diabetes and xerostomia

- Xerostomia and hyposalivation is frequent among diabetes mellitus patients.
- **Etiology:** damage to salivary gland parenchyma, alterations in salivary gland microcirculation, dehydration, and mitochondrial dysfunction.
- **Complications:** caries, damage to the gingiva and supporting tissues is more extensive in diabetic patients compared to non-diabetic patients and increase risk for infection.

[Type 2 diabetes-induced hyposalivation of the submandibular gland through PINK1/Parkin-mediated mitophagy - PMC \(nih.gov\)](#).

[Periodontal disease and diabetes mellitus: the role of tumor necrosis factor-alpha in a 2-way relationship - PubMed \(nih.gov\)](#)

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## Diabetes and Periodontal disease

### Two-way relationship

- Damage to the gingiva and supporting tissues is more extensive in diabetic patients. Patients with uncontrolled diabetes have increased risk for periodontitis as well as bone loss.
- Treating periodontal diseases may improve glycemic control. Uncontrolled periodontal disease can lead to poor glycemic control.
- Mechanism is not well elucidated. Upregulated inflammatory cytokines and immune dysregulation may contribute to the two-way relation.

# **Hyperglycemia in pregnancy & Diabetes Insipidus**

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**Gestational diabetes mellitus (GDM)** is a transient form of diabetes that occurs during pregnancy due to increased insulin resistance. In uncomplicated cases, hyperglycemia resolves during the postpartum period. GDM can increase the risk Type 2 diabetes. More common (80 %).

**Overt Diabetes** is hyperglycemia recognized during pregnancy. Meets diagnostic threshold of diabetes in non-pregnant adults. More severe form of hyperglycemia and is associated with worse maternal and fetal outcomes. Less common (15-20 %).

### **Pathophysiology:**

The exact mechanism of gestational diabetes is unknown.

- In normal pregnancy, insulin sensitivity usually declines with advancing gestation, but euglycemia is maintained by increased insulin secretion.
- In GDM, insulin resistance , inadequate insulin secretion and pregnancy hormones are contributing factors.

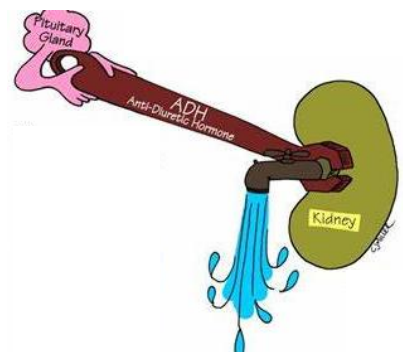
### **Complications:**

Hyperglycemia and excessive fetal growth, which can increase the risk of maternal morbidity, 3<sup>rd</sup> trimester pregnancy loss/stillbirth and fetal complications.

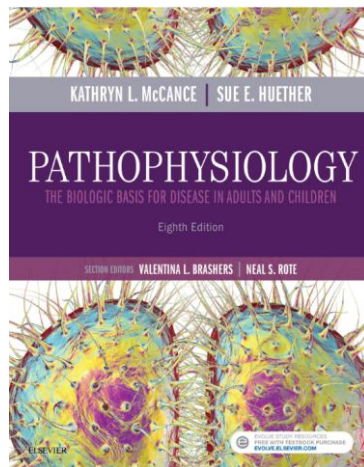
## Diabetes Insipidus

Diabetes insipidus is characterized by excretion of abnormally large volumes of dilute urine without elevated blood sugar level. Symptoms: polyuria (2–20 L/d) and polydipsia

- **Central diabetes insipidus**: due to deficient secretion of anti-diuretic hormone (ADH).
- **Nephrogenic diabetes insipidus**: due to renal resistance to the anti-diuretic hormone.



## Resources



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